

Sangjae Lee

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RESEARCH INTERESTS

Wireless sensing, indoor localization, geomagnetic positioning, sensor fusion, and AI-driven network intelligence for real-world indoor environments.

EXPERIENCE

- **Samsung Research** Apr 2022 – Present
Staff Engineer Seoul, Korea
 - Contributed to indoor localization systems using geomagnetic signals, Wi-Fi, and conventional machine learning approaches
 - Developed learned sensing and inference models for same-space discrimination, motion-based nearby device detection, and context-aware cellular/Wi-Fi network switching
 - Developing an LLM-based network optimization agent for decision support in wireless environments

EDUCATION

- **Korea Advanced Institute of Science and Technology** Sep 2012 – Feb 2021
PhD in Computer Science Daejeon, Republic of Korea
 - Advisor: Prof. Dongsoo Han
 - Dissertation: “High-precision Indoor Positioning using Geomagnetic Anomaly”
 - Committee: Dongsoo Han (Chair), Giwan Yoon, Geehyuk Lee, Junekoo Rhee, and Hojin Choi
- **Korea Advanced Institute of Science and Technology** Sep 2010 – Aug 2012
MS in Computer Science Daejeon, Republic of Korea
 - Advisor: Prof. Dongsoo Han
 - Thesis: “Missing signal imputation for accurate WLAN-based indoor localization system”
- **Hanyang University** Mar 2006 – Feb 2010
BS in Computer Science Seoul, Republic of Korea
 - Project: “Intelligent Lane Detection”
 - LEE GAHYUNG scholarship, 3.91/4.5 (Dean’s List)

SELECTED PUBLICATIONS(J=JOURNAL, C=CONFERENCE)

- [J.4] Lee, S., Chae, S., & Han, D. (2020). **Indoor Localization Using Augmented Vector of Geomagnetic Field**. *IEEE Access*.
- [J.3] Lee, G., Moon, B.-C., Lee, S., & Han, D. (2020). **Fusion of the SLAM with Wi-Fi-Based Positioning Methods for Mobile Robot**. *Sensors*.
- [C.2] Lee, S., & Han, D. (2019). **Rover Who Make Indoor Radio Map**. *MobiSys 2019*.
- [C.1] Lee, S., et al. (2017). **Construction of an Indoor Positioning System for Home IoT Applications**. *ICC 2017*.
- [J.2] Jung, S.-h., Lee, S., & Han, D. (2016). **A Crowdsourcing-Based Global Indoor Positioning and Navigation System**. *Pervasive and Mobile Computing*.
- [J.1] Han, D., Jung, S.-h., & Lee, S. (2016). **A Sensor Fusion Method for Wi-Fi-Based Indoor Positioning**. *ICT Express*.

RESEARCH EXPERIENCE

- **Huawei R&D** Oct 2018 – Sep 2021
Automated Techniques for Crowdsourcing Radio Map Construction Shanghai, China
 - Industry-supported research on scalable radio map construction for indoor localization
- **National Research Foundation of Korea** 2019 – 2021
Crowdsourcing-based Indoor GPS System and GIPS-based Science Museum Services Daejeon, Korea
 - Projects: Crowdsourcing-based Indoor GPS System (Mar 2020 – Feb 2021); Integrated Operation Technology for Science Museum Services Based on GIPS (Jan 2019 – Dec 2019)
- **Korea Advanced Institute of Science and Technology** 2016 – 2020
Hospital infection prevention and robot-based indoor positioning research Daejeon, Korea
 - Projects: COVID-19 Infection Prevention System in Hospitals (Aug 2020 – Dec 2020); Robot-based Indoor Positioning for Location-based Home IoT Service Platform (Sep 2016 – Dec 2016)

- **Samsung Electronics** 2013 – 2018
Indoor positioning, radio map generation, and malicious AP detection Suwon, Korea
 - Projects: Location-based Malicious Access Point Detection Technology (May 2018 – Dec 2018); Wi-Fi-based Map Construction and Positioning Technology (Apr 2016 – Mar 2017); Automatic Generation and Update Technology for Radio Maps (May 2013 – Feb 2014)

PROJECTS

- **KAILOS | Crowdsourcing indoor positioning service platform** Jun 2014 – 2016
Backend and platform development kailos.io
 - Contributed to backend APIs, data ingestion workflows, map management, and positioning services for a public crowdsourced indoor localization platform
- **IPIN'14 | On-site Indoor Positioning and Navigation Competition** 2014
Scenario design and operations Busan, Korea
 - Designed competition scenarios and supported on-site operational management

SKILLS

- **Languages:** Python, Java, JavaScript, MATLAB, R, C, Kotlin
- **Modeling & Analysis:** Deep learning, conventional machine learning, sensor fusion, data processing
- **Systems & Domains:** Indoor localization, wireless sensing, radio map construction, network optimization